

Question:

A 13-year-old male is admitted to the hospital for treatment of acute lymphoblastic leukemia. During his hospital course, he develops a fever of 39.0 degrees Celsius. A CBC demonstrates a leukocyte count of $<500 /\text{mm}^3$. Which of the following is the most appropriate initial management of this patient? Answer Choices: A. Granulocyte colony-stimulating factor (G-CSF) B. IV ceftazidime C. Oral doxycycline D. Oral ciprofloxacin and amoxicillin/clavulanic acid

Reasoning + Answer:

<think>The patient is a pediatric acute lymphoblastic leukemia patient with a very low leukocyte count and a fever, which could indicate a severe neutropenic fever (febrile neutropenia), a complication in cancer patients with a low white blood cell count that can be life-threatening due to the high risk of infection, and the most common initial treatment for this is to use a hematopoietic growth factor like granulocyte colony-stimulating factor (G-CSF) to stimulate white blood cell production and help combat the infection.</think><answer> A . Granulocyte colony-stimulating factor (G-CSF) </answer>

